Netcat (nc) Cheat Sheet

TCP, UDP, ports, listeners, file transfer, HTTP, Ncat and troubleshooting

Note: Options vary between Netcat versions. Check your local documentation. Test only systems you own or are authorized to assess.

1. Common Flags

Flag	Purpose
`-l`	Listen for connections
`-v`, `-vv`	Verbose output
`-n`	Disable DNS lookups
`-z`	Test TCP connectivity
`-u`	Use UDP
`-w N`	Timeout, where supported
`-k`	Keep listener open, where supported
`-4 / -6`	Use IPv4 / IPv6, where supported

```
nc [options] HOST PORT
nc -l PORT
```

2. TCP Client and Listeners

```
# Connect to a service
nc example.com 80
nc -nv -w 5 192.168.1.20 443

# Terminal 1: local listener
nc -lv 4444

# Terminal 2: connect
nc 127.0.0.1 4444

# Ncat listener
ncat -lk 4444
```

Type messages between terminals. Use loopback for local practice; stop listeners with Ctrl+C.

3. TCP Port Testing

```
# Test one port
nc -vz -w 3 HOST 443

# Test common ports individually
nc -vz HOST 22
nc -vz HOST 80
nc -vz HOST 443

# Test a local service
nc -vz 127.0.0.1 8080
```

4. UDP Testing

```
# UDP listener
nc -ul 5000

# UDP client
nc -u 127.0.0.1 5000

# Send a test message
printf 'Hello\n' | nc -u -w 2 127.0.0.1 5000
```

UDP is connectionless. Sending data does not guarantee delivery or a response.

A successful connection confirms TCP connectivity at that moment. A timeout may indicate filtering, routing issues or an unavailable host; it does not prove a port is closed. Use Nmap for authorized port scanning.

5. Send Text and Data

```
# Send a message
printf 'Hello\n' | nc -w 3 HOST PORT

# Send multiple lines
printf 'First\nSecond\n' | nc -w 3 HOST PORT

# Send a file
nc HOST PORT < input.txt

# Receive data into a file
nc HOST PORT > output.bin

# Send command output
some_command | nc HOST PORT
```

6. File Transfer

```
# Receiver
nc -lv 9000 > received.bin

# Sender
nc RECEIVER_IP 9000 < original.bin

# Verify integrity (Linux)
sha256sum original.bin received.bin

# macOS
shasum -a 256 original.bin received.bin

# PowerShell
Get-FileHash .\original.bin -Algorithm SHA256
Get-FileHash .\received.bin -Algorithm SHA256
```

> Transfers are unencrypted and unauthenticated. Use SSH/SCP/SFTP for confidential files, and never blindly extract untrusted archives.

7. HTTP Testing

```
# Connect
nc example.com 80

# Send an HTTP GET request
printf 'GET / HTTP/1.1\r\nHost: example.com\r\nConnection:
close\r\n\r\n' |
nc -w 5 example.com 80

# Send a HEAD request
printf 'HEAD / HTTP/1.1\r\nHost: example.com\r\nConnection:
close\r\n\r\n' |
nc -w 5 example.com 80
```

Include the blank line ending the HTTP headers. HTTPS requires TLS; use curl or a TLS-capable Ncat build.

8. Ncat Essentials

Task	Command
Listen	`ncat -l 4444`
Keep listener open	`ncat -lk 4444`
Timeout	`ncat -w 5 HOST PORT`
TLS client	`ncat --ssl HOST 443`
Log session	`ncat -o session.log HOST PORT`
Choose source port	`ncat --source-port 5555 HOST PORT`
Help	`ncat --help`

Options and TLS verification behavior depend on version and configuration.

9. Network Diagnostics

Task	Linux	macOS / Windows
IP address	`ip addr`	`ifconfig` / `ipconfig`
Routes	`ip route`	`netstat -rn` / `route print`
TCP listeners	`ss -lnt`	`lsof -nP -iTCP -sTCP:LISTEN` / `netstat -ano`
UDP listeners	`ss -lnu`	`netstat -ano -p udp` (Windows)
DNS lookup	`dig HOST`	`nslookup HOST`

Reachability	`ping HOST`	`ping HOST`
Connections	`ss -tn`	`netstat -an`

10. Bash Automation

```
# Check exit status
nc -z -w 3 127.0.0.1 8080
echo $?

# Conditional check
if nc -z -w 3 127.0.0.1 8080; then
    echo "Port reachable"
else
    echo "Port not reachable"
fi

# Check selected local ports
for port in 22 80 443 8080; do
    if nc -z -w 2 127.0.0.1 "$port"; then
        echo "$port reachable"
    else
        echo "$port unavailable"
    fi
done
```

11. Common Service Ports

Port	Service
20–21	FTP
22 / 23	SSH / Telnet
25, 465, 587	SMTP
53 / 67–68	DNS / DHCP
80 / 443	HTTP / HTTPS
110, 143, 993, 995	POP3 / IMAP and TLS variants
123 / 161	NTP / SNMP
389 / 636	LDAP / LDAPS
445	SMB
1433 / 1521	SQL Server / Oracle DB
3306 / 5432	MySQL / PostgreSQL
3389 / 5900	RDP / VNC
6379 / 27017	Redis / MongoDB
8080 / 8443	Alternative HTTP / HTTPS

These are common assignments, not guarantees of which service is running.

12. Troubleshooting

Problem	What to check
Command not found	Install Netcat or Ncat.
Invalid option	Check the installed version's help.
Connection refused	Verify listener and destination port.

13. Security Checklist

- Test only authorized systems.
- Prefer loopback for local experiments.
- Stop listeners when finished and restrict firewall access.
- Never send credentials over plaintext connections.

Timeout	Check firewall, routing and filtering.
Address already in use	Find the existing listener or choose another port.
Local works, remote fails	Check bind address and firewall rules.
UDP gets no response	Verify service, destination and return path.
Empty received file	Check sender input and connection closure.
HTTPS fails	Use a TLS-capable client.

- Use SSH/SFTP for confidential transfers.
- Never expose unauthenticated shell access.
- Do not blindly execute received data or extract untrusted archives.

14. Quick Reference

```
# TCP client / listener
nc HOST PORT
nc -lv 4444

# Test TCP port
nc -vz -w 3 HOST 443

# UDP client / listener
nc -u HOST PORT
nc -ul 5000

# Send text
printf 'Hello\n' | nc -w 3 HOST PORT

# Transfer file
nc -lv 9000 > received.bin
nc HOST 9000 < original.bin

# Inspect listeners
ss -lnt

# Ncat TLS client
ncat --ssl HOST 443
```

15. Official Documentation

- OpenBSD Netcat Manual
- Ncat User's Guide
- Ncat Usage
- Nmap Reference Guide